

- 1) A K-band police radar operates at a frequency of 24.15 GHz. Calculate the period of the wave, the wavelength in air, and the wavenumber of the wave in air.

$$T = 1/f = (24.15 \text{ E } 9)$$

$$T = 4.14 \text{ E } -11$$

Wave length in the air:

$$\lambda_0 = \frac{c}{f} = \frac{3 \text{ E } 8}{24.15 \text{ E } 9} = 1.242 \text{ E } -2 \text{ [m]}$$

$$\text{Wavenumber in air} = \frac{2\pi}{\lambda_0} = \frac{2\pi}{1.242 \text{ E } -2} = 505.8 \text{ [rad/m]}$$

- 2) A "dipole" antenna consists of a metal wire that is fed at the center. The dipole antenna is optimum in receiving signals when its length is approximately equal to one-half the wavelength for the signal of interest. What are optimum dipole antenna lengths for receiving signals for the following TV stations:

(a) Channel 8 ($f = 181.25 \text{ MHz}$)

(b) Channel 26 ($f = 543.25 \text{ MHz}$)

$$a) \lambda_0 = \frac{c}{f} \Rightarrow 2l = \frac{c}{f} \Rightarrow \frac{c}{2f}$$

$l = \text{length}$

$$l = \frac{3 \text{ E } 8}{2(181.25 \text{ E } 6)} = 0.828 \text{ [m]}$$

$$b) l = \frac{3 \text{ E } 8}{2(543.25 \text{ E } 6)} = 0.276 \text{ [m]}$$

- 3) A uniform plane wave has an electric field vector that is given by

$$\underline{E}(x, y, z, t) = (120\hat{x} - 50\hat{y}) \cos(10^9 t - kz) \text{ [V/m]}.$$

This plane wave is propagating in polypropylene ($\epsilon_r = 2.0$), which may be assumed lossless and non-magnetic. Find the following (and make sure that you include units):

(a) k ; (b) λ ; (c) $\underline{E}(x, y, z)$; (d) $\underline{H}(x, y, z)$; (e) $\underline{H}(x, y, z, t)$

$$\text{Given: } \underline{E}(x, y, z, t) = (120\hat{x} - 50\hat{y}) \cos(10^9 t - kz) \text{ [V/m]}$$

$\epsilon_r = 2.0$, lossless, nonmagnetic ($\mu_r = 1$)

$$(a) k = k_0 \sqrt{\epsilon_r} = \frac{\omega}{c} \sqrt{\epsilon_r} = \frac{10^9}{3 \text{ E } 8} \sqrt{2.0}$$

$$k = 4.714 \text{ [rad/m]}$$

(b) Wave length λ

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{4.714} = 1.333 \text{ [m]}$$

(c) Phasor $\underline{E}(x, y, z)$

$$\underline{E}(x, y, z) = (120\hat{x} - 50\hat{y}) e^{-jkz} \text{ [V/m]}$$

(d) Phasor $\underline{H}(x, y, z)$, for a wave traveling in $+z$:

$$\underline{H} = \frac{\hat{z} \times \underline{E}}{\eta}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{376.73}{\sqrt{2.0}} = 266.4 \text{ [}\Omega\text{]}$$

$$\text{Now compute } \hat{z} \times \underline{E}: \hat{z} \cdot (120\hat{x} - 50\hat{y}) = 120(\hat{z} \times \hat{x}) - 50(\hat{z} \times \hat{y})$$

$$= 120\hat{y} + 50\hat{x} = 50\hat{x} + 120\hat{y}$$

$$\underline{H}(x, y, z) = \frac{(50\hat{x} + 120\hat{y})}{266.4} e^{-jkz} = (0.1877\hat{x} + 0.4504\hat{y}) e^{-jkz} \text{ [A/m]}$$

$$e) \underline{H}(x, y, z, t) = (0.1877\hat{x} + 0.4504\hat{y}) \cos(10^9 t - kz) \text{ [A/m]}$$

- 4) A certain electromagnetic wave that is in a vacuum has frequency f_0 , wavelength λ_0 , wavenumber k_0 , and phase velocity v_p . When it enters a nonmagnetic dielectric medium characterized by $\epsilon_r = 2.56$, what are the new values of f , λ , k , and v_p for the wave in this medium (in terms of the values for the same wave in vacuum)?
- 5) A lossy nonmagnetic dielectric has $\epsilon_r = 2.0$ and $\sigma/(\omega\epsilon) = 2.0 \times 10^{-4}$.
(Since the relative permittivity is real, note that this material has no polarization loss, but only conducting loss.) For a frequency of 250 [MHz], how far can a wave propagate in the material before (a) it undergoes an attenuation of 1 [Np]; (b) it undergoes an attenuation of 10 dB; (c) its amplitude is halved; (d) its phase shifts 180° .
- 6) The ocean has a relative permittivity of $\epsilon_r = 81$ and a conductivity of $\sigma = 4.0$ [S/m]. The ocean water may be assumed to be nonmagnetic. Determine the dB of attenuation for a plane wave that propagates from the surface of the ocean to a depth of 100 [m], if the frequency is 750 [Hz].
- 7) Calculate the skin depth and the surface impedance for aluminum at a frequency of 2.45 [GHz] (this is the frequency of a microwave oven). The conductivity of aluminum is taken as 2.0×10^7 [S/m]. Aluminum is nonmagnetic ($\mu = \mu_0$).

- 4) A certain electromagnetic wave that is in a vacuum has frequency f_0 , wavelength λ_0 , wavenumber k_0 , and phase velocity v_p . When it enters a nonmagnetic dielectric medium characterized by $\epsilon_r = 2.56$, what are the new values of f , λ , k , and v_p for the wave in this medium (in terms of the values for the same wave in vacuum)?

$\epsilon_r = 2.56$ for non magnetic mediums the relative permeability can be assumed to be 1. $\mu_r = 1$

$$\lambda_g = \lambda_d = \frac{\lambda_0}{\sqrt{\epsilon_r \mu_r}}$$

a) there is no change in frequency, so $f = f_0$

where

$$c_d = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

$$k = \omega \sqrt{\mu \epsilon} = k_0 \sqrt{\mu_r \epsilon_r} \quad (\text{wavenumber of dielectric medium})$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\mu = \mu_0 \mu_r$$

14

$$\lambda_0 = \frac{c}{f}$$

$$v_p = c_d$$

b) $\lambda_d = \frac{\lambda_0}{\sqrt{2.56(1)}} = 0.625 \lambda_0$

c) $k = \omega \sqrt{\mu \epsilon} = k_0 \sqrt{\mu_r \epsilon_r}$

$$K = k_0 \sqrt{\mu_r \epsilon_r} = 1.6 k_0$$

d) $v_{\text{phase}} : \text{if } v_p = c_d, \text{ then } v_p = \frac{1}{\sqrt{\epsilon_r \mu_r}} v_{p0} = 0.625 v_{p0}$

- 5) A lossy nonmagnetic dielectric has $\epsilon_r = 2.0$ and $\sigma/(\omega \epsilon) = 2.0 \times 10^{-4}$.

(Since the relative permittivity is real, note that this material has no polarization loss, but only conducting loss.) For a frequency of 250 [MHz], how far can a wave propagate in the material before (a) it undergoes an attenuation of 1 [Np]; (b) it undergoes an attenuation of 10 dB; (c) its amplitude is halved; (d) its phase shifts 180° .

15.8.2 Low-Loss Limit

When $\tan \delta \ll 1$, approximate formulas simplify the analysis:

$$\alpha \approx \frac{k \tan \delta}{2}, \quad \beta \approx k = \omega \sqrt{\mu \epsilon'}$$

$$\epsilon_r = 2.0 \quad \tan \delta = 2.0 \times 10^{-4}$$

$$\delta = \frac{\sigma}{(\omega \epsilon_0 \epsilon_r)} = 2 \times 10^{-4} \quad 2.0 \times 10^{-4} \ll 1$$

$$f = 250 \text{ MHz} \quad \omega = 500\pi \text{ rad/s}$$

$$\alpha = k'' \quad \beta = k'$$

$$k' = \frac{2\pi}{\lambda_g} \quad k'' = 1/d_p$$

i.) using the low loss approximation

$$d_p \approx \sqrt{\frac{\epsilon}{\mu_0}} \left(\frac{2}{\sigma} \right) \quad (\tan \delta \ll 1)$$

$$\alpha = k'' = \frac{1}{d_p} = \frac{\sigma}{2} \sqrt{\frac{\epsilon}{\mu_0}} \Rightarrow \frac{k_0 \sqrt{\epsilon_r}}{2} \tan \delta = \frac{\omega \sqrt{\mu \epsilon}}{2} \cdot \frac{\sigma}{\omega \epsilon}$$

$$\beta = k' = \omega \sqrt{\mu \epsilon} = k_0 \sqrt{\epsilon_r}$$

$$k_0 = \frac{2\pi f}{c} = 5.2396 \text{ [rad/m]}$$

$$\epsilon_r = 2.0$$

$$\beta = 5.2396 \sqrt{2} = 7.4099 \text{ [rad/m]}$$

$$\alpha = \frac{\beta}{2} \cdot \frac{\sigma}{\omega \epsilon} = \frac{7.4099}{2} \cdot 2.0 \times 10^{-4} = 7.4099 \times 10^{-4} \text{ [Np/m]}$$

$$\alpha = 7.4099 \times 10^{-4} \text{ [Np/m]}$$

$$\beta = 7.4099 \text{ [rad/m]}$$

The "depth of penetration" d_p is defined.

$$k'' d_p = 1 \Rightarrow d_p \equiv 1/k''$$

a) $\alpha z = 1 \text{ Np} \Rightarrow z = \frac{1}{\alpha} = \frac{1}{7.4099 \times 10^{-4}}$

$$z = 1349.54 \text{ m}$$

b) $10 \log_{10}(e^{-2\alpha z}) = -10 \text{ dB}$

$$2\alpha z \cdot 10 \log_{10}(e) = -10$$

$$2\alpha z \cdot 4.343 = 10$$

$$z = \frac{10}{8.686 \cdot \alpha} = \frac{10}{8.686 \cdot 7.4099 \times 10^{-4}}$$

$$z = 1553.7 \text{ [m]}$$

c) $e^{-\alpha z} = 0.5 \Rightarrow \ln e^{-\alpha z} = \ln(0.5) \Rightarrow -\alpha z = \ln(0.5) = -\ln(2)$

$$z = \frac{\ln(2)}{\alpha} = 935.43 \text{ m}$$

d) $\beta z = \pi \Rightarrow z = \frac{\pi}{\beta}$

$$z = 0.424 \text{ [m]}$$

- 6) The ocean has a relative permittivity of $\epsilon_r = 81$ and a conductivity of $\sigma = 4.0$ [S/m]. The ocean water may be assumed to be nonmagnetic. Determine the dB of attenuation for a plane wave that propagates from the surface of the ocean to a depth of 100 [m], if the frequency is 750 [Hz].

given: $\epsilon_r = 81$, $\sigma = 4.0$ [S/m], $\mu_r = 1$, $f = 750$ [Hz], $z = 100$ [m]

$$\tan \delta = \frac{\sigma}{\omega \epsilon} = \frac{\sigma}{\omega \epsilon_r \epsilon_0} = \frac{4}{2\pi(750)(81)(8.854 \times 10^{-12})}$$

$\tan \delta = 1.184 \times 10^6$, $\tan \delta > 1$, this is a good conductor

Find Skin Depth:

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{2}{2\pi(750)(1.2566 \times 10^{-6})(4)}} = 9.187 \text{ [m]}$$

Find α , $K'' = \frac{1}{\delta} = \frac{1}{9.187} = 0.1088$ [Np/m]

$z = 100$ [m]

$= 8.686 \cdot \alpha \cdot z = 8.686 \cdot 0.1088 \cdot 100 = 94.5 \text{ dB}$

Attenuation = 94.5 dB

- 7) Calculate the skin depth and the surface impedance for aluminum at a frequency of 2.45 [GHz] (this is the frequency of a microwave oven). The conductivity of aluminum is taken as 2.0×10^7 [S/m]. Aluminum is nonmagnetic ($\mu = \mu_0$).

$f = 2.45 \times 10^9$ [Hz]

$\omega = 2\pi f$

$\sigma = 2.0 \times 10^7$ [S/m]

$\omega = 2\pi \cdot 2.45 \times 10^9$

$\mu = \mu_0 = 4\pi \times 10^{-7}$ [H/m]

Skin Depth

$$\delta_s = \sqrt{\frac{2}{\omega \mu \sigma}} \text{ [m]}. \quad (16.2)$$

The fields decay as e^{-z/δ_s} and oscillate as e^{-jz/δ_s} . The attenuation and phase constants are equal: $\alpha = \beta = 1/\delta_s$.

$\mu = \mu_0 = 4\pi \times 10^{-7}$ [H/m]

skin depth: $\delta_s = \sqrt{\frac{2}{\omega \mu \sigma}} \text{ [m]} = \sqrt{\frac{2}{(2\pi)(2.45 \times 10^9)(4\pi \times 10^{-7})(2 \times 10^7)}} = 2.27 \times 10^{-6} \text{ [m]}$

surface impedance: $Z_s = \left(\frac{1}{\sigma \delta}\right)(1+j) \text{ [\Omega]} = \frac{1}{((2.0 \times 10^7)(2.27 \times 10^{-6}))} \cdot (1+j) = 0.0212 + 0.0212j$

Surface Impedance (cont.)

$$Z_s = \left(\frac{1}{\sigma \delta}\right)(1+j) \text{ [\Omega]}$$

Define "surface resistance" and "surface reactance" from: $Z_s = R_s + jX_s$

We then have:

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\omega \mu}{2\sigma}}$$

$$X_s = R_s$$

$$Z_s = R_s(1+j) \text{ [\Omega]}$$

Recall:

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}}$$

- 8) A sheet of aluminum foil has a thickness of about 1 mil (a mil is 0.001 inches). Calculate the dB of attenuation of a plane wave that goes through the sheet of aluminum foil, at a frequency of 2.45 [GHz]. Is the sheet of aluminum foil a good shield for the microwaves that are inside a microwave oven operating at this frequency?

- 9) Find the polarization (linear, circular, or elliptical) and handedness (left-handed or right-handed) for the following fields:

(a) $\underline{E} = (\underline{\hat{x}} - j\underline{\hat{y}})e^{-jkz}$

(b) $\underline{E} = [(3 - j2)\underline{\hat{x}} + (1 + j)\underline{\hat{z}}]e^{-jky}$

(c) $\underline{E} = [j3\underline{\hat{y}} - j2\underline{\hat{z}}]e^{jkx}$

(d) $\underline{E} = [(1 + j)\underline{\hat{x}} + (1 - j)\underline{\hat{y}}]e^{jkz}$

- 10) A plane wave that is propagating in the positive z direction has the following electric field at $z = 0$: $\underline{E} = [\underline{\hat{x}}(2.5)\cos(\omega t) + \underline{\hat{y}}(1.5)\sin(\omega t - 10^\circ)]$. Determine what type of wave this is (RHEP, RHCP, LHEP, LHCP, linear), using the graphical rotation rule.

- 8) A sheet of aluminum foil has a thickness of about 1 mil (a mil is 0.001 inches). Calculate the dB of attenuation of a plane wave that goes through the sheet of aluminum foil, at a frequency of 2.45 [GHz]. Is the sheet of aluminum foil a good shield for the microwaves that are inside a microwave oven operating at this frequency?

Thickness: 1 mil = 0.001 inches = $2.54 \times 10^{-5} = 25.4 \times 10^{-6}$ [m]

$$\delta = 2.27 \times 10^{-6} \text{ [m]}$$

$$\frac{1}{\delta} = \frac{1}{2.27 \times 10^{-6}} = 4.406 \times 10^{-5} \text{ [NP/m]}$$

Attenuations = $(8.686)(\alpha)(t) = (8.686)(4.406 \times 10^{-5})(25.4 \times 10^{-6}) = 97.2 \text{ dB}$

$\frac{t}{\delta} = \frac{25.4}{2.27} = 11.2$ skin depth, the foil is an excellent shield

- 9) Find the polarization (linear, circular, or elliptical) and handedness (left-handed or right-handed) for the following fields:

(a) $\underline{E} = (\hat{x} - j\hat{y})e^{-jkz}$

(b) $\underline{E} = [(3 - j2)\hat{x} + (1 + j)\hat{z}]e^{-jky}$

(c) $\underline{E} = [(j3\hat{y} - j2\hat{z})]e^{jkx}$

(d) $\underline{E} = [(1 + j)\hat{x} + (1 - j)\hat{y}]e^{jkz}$

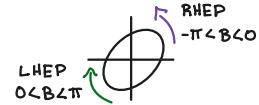
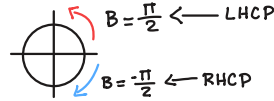
The Key rules:

linear: Components are in phase or 180° out of phase

Circular: equal amplitude, 90° phase difference

Elliptical: everything else

Handedness: use right hand rule



a) $\underline{E} = (\hat{x} - j\hat{y})e^{-jkz}$

$x: \sqrt{1} = 1$ $y: \sqrt{1} = 1$ normalized: $\epsilon_y = j/1 = -j = -\pi/2$
same magnitude = circular

right-hand Circular Polarization

b) $\underline{E} = [(3 - j2)\hat{x} + (1 + j)\hat{z}]e^{-jky}$

$\epsilon_x = \sqrt{3^2 + 2^2} = \sqrt{13}$ $\epsilon_y = \sqrt{1^2 + 1^2} = \sqrt{2}$ ← Elliptical polarization

normalize: $\frac{1+j}{3-j2} = \frac{\sqrt{2}}{\sqrt{13}} = 0.39223 < 78.6901$ ← LHEP

c) $[(j3\hat{y} - j2\hat{z})]e^{jkx}$

$\sqrt{3^2} = 3 = \frac{\pi}{2}$ $\sqrt{2^2} = 2 = -\frac{\pi}{2}$

$\beta = \text{Phase of } y - \text{phase } z = 90^\circ + 90^\circ = \pi$ linear Polarization

d) $\underline{E} = [(1 + j)\hat{x} + (1 - j)\hat{y}]e^{jkz}$

$\sqrt{1^2 + 1^2} = \sqrt{2}$ $\sqrt{1^2 + 1^2} = \sqrt{2}$

normalize: $\frac{1-j}{1+j} = 1 \angle -90^\circ$ RHCP

- 10) A plane wave that is propagating in the positive z direction has the following electric field at $z = 0$: $\underline{E} = [\hat{x}(2.5)\cos(\omega t) + \hat{y}(1.5)\sin(\omega t - 10^\circ)]$. Determine what type of wave this is (RHEP, RHCP, LHEP, LHCP, linear), using the graphical rotation rule.

$a = 2.5$ $b = 1.5$ $B = -100^\circ$

We know $B = -100 \rightarrow -\pi < B < 0 \Rightarrow \text{RHEP}$